

RAHAT AHMED JOBU

PhD Applicant: Machine Learning, NLP & Software Security

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Research Profile

CSE student at BRAC University working toward a PhD in machine learning, NLP, and software security. My research centers on detecting systems designed to look trustworthy but aren't: fraudulent text and job postings, malicious software packages and IDE extensions, silent API breaking changes, and LLM reliability across languages. A defining practice across this work is adversarial self-evaluation: multiple rounds of self-directed critique targeting train/test leakage, confounded baselines, and unverified claims, applied to my own results before they reach peer review.

Qualifications

Bachelor of Science in Computer Science

BRAC University, Dhaka | Expected Graduation: December 2026

Higher Secondary Certificate (HSC), Science: GPA 5.00/5.00

Shaheed Police Smrity College

Secondary School Certificate (SSC), Science: GPA 5.00/5.00

Green Heart School

Publications

Peer-Reviewed Journal Article (Published)

Jobu, R. A. "An Explainable, Cost-Aware Hybrid Framework for Fraudulent Online Job Posting Detection." **Array** (Elsevier), **Q1 Journal**, 2026. **First & Corresponding Author.**

Working Papers & Manuscripts Under Review

- Engineering a Lightweight, Cross-Generator-Robust Pipeline for Bengali AI-Paraphrase Detection: A Leakage- and Artifact-Aware Evaluation Process. Target: *Results in Engineering*. Under Review.
- Does Naive Cross-Validation Inflate Apparent Performance in Software Energy-Regression Prediction? Target: *Information and Software Technology*. Under Review.

Book

Predicting Silent Behavioral Drift Severity: From Default-Argument Breaking Changes in Python Machine Learning Libraries. Available on Amazon.

Appointments and Experience

Software Engineer

ASG SHOP™ (AI Solutions Garage) | Jul 2026 – Present

- Design, develop, and maintain features across the company's software products; collaborate with cross-functional teams to translate business requirements into technical solutions.
- Part of the engineering team maintaining the technical infrastructure behind Apar's Classroom, an EdTech

platform serving 1M+ students in Bangladesh, ranked #2 globally on TIME's World's EdTech Rising Stars 2026 list.

Software Engineering Intern

ASG SHOP™ (AI Solutions Garage) | Jan 2026 – Jun 2026

- Assisted in the design, development, and testing of web application modules under senior engineers' guidance; participated in code reviews, daily stand-ups, and sprint planning.

Software Engineer

IOOTE | Dec 2024 – Sep 2025

- Developed full-stack web features using React, Node.js, and MongoDB for tech-gadget products.
- Built REST APIs and integrated front-end and back-end services; collaborated with designers and product stakeholders to deliver production-ready features.

Selected Projects

- **Job Fraud Detector:** Hybrid SVM + DistilBERT pipeline for fraudulent online job posting detection on the EMSCAD dataset; implementation behind the published *Array* paper. github.com/rahat239/job-fraud
- **Malware Classifier (PyPIGuard):** ML tool for pre-install detection of malicious PyPI packages, using cluster-disjoint evaluation to eliminate near-duplicate leakage. github.com/rahat239/PyPIguard-Research
- **Techra:** PC Builder & E-Commerce Platform (MERN stack). github.com/rahat239/TechRA
- **J.A.R.V.I.S.:** Personal macOS AI assistant (~5,500 lines of Python) with voice, gesture, and face recognition

Entrepreneurship

Founder, MeowMatchBD (meowmatchbd.com)

A full-stack cat ecosystem platform for Bangladesh: adoption listings for breed cats, a live rescue system, and nearby cat and veterinarian discovery. Revenue generated through paid sell postings. Grown to 1,200+ registered users and 218 cats adopted to date. Responsible for end-to-end product development, from initial build through ongoing operations.

Achievements

- Selected among Top 8 teams at University Project Showcasing Event, evaluated by industry experts from Brain Station 23, Optimizely, and Cefalo Bangladesh, recognizing technical innovation, system design, and practical applicability.

Technical Skills

Machine Learning / NLP: Python, PyTorch, Scikit-learn, Transformers, Statistical Evaluation (cluster-disjoint validation, bootstrap resampling)

Also proficient in: JavaScript, React, Node.js, Express.js, MongoDB, C++, Arduino, Git